

FACE: ArcFace White-box Geometric Identity Optimization

Frozen iResNet-100 identity-distance results with downstream InstructPix2Pix evaluation

Author: Parth Katiyar

FACE optimizes $Z = 1 - \text{cosine_similarity}$ between original and perturbed full-image ArcFace iResNet-100 embeddings. The optimized loss is exactly $\text{loss} = -Z$. ArcFace weights are frozen; only geometry parameters are optimized.

No landmarks, face detection, alignment, visual counter-loss, VAE objective, or UNet objective is used. Prompts are retained only for downstream InstructPix2Pix evaluation.

Run matrix and final values

face	prompt	final Z	best Z	cosine sim	score %	SSIM	PSNR	edit SSIM	max disp
face_002	add black sunglasses	0.0592	0.0593	0.9408	94.081	0.9737	24.621	0.6466	6.3730
face_002	add headphones	0.0680	0.0681	0.9320	93.198	0.9733	24.551	0.6269	7.2244
face_005	add black sunglasses	0.1030	0.1032	0.8970	89.704	0.9619	22.976	0.6888	18.171
face_005	add headphones	0.0990	0.0991	0.9010	90.102	0.9613	22.908	0.6790	11.920

Case image strips

face_002 / add black sunglasses



face_002 / add headphones



face_005 / add black sunglasses

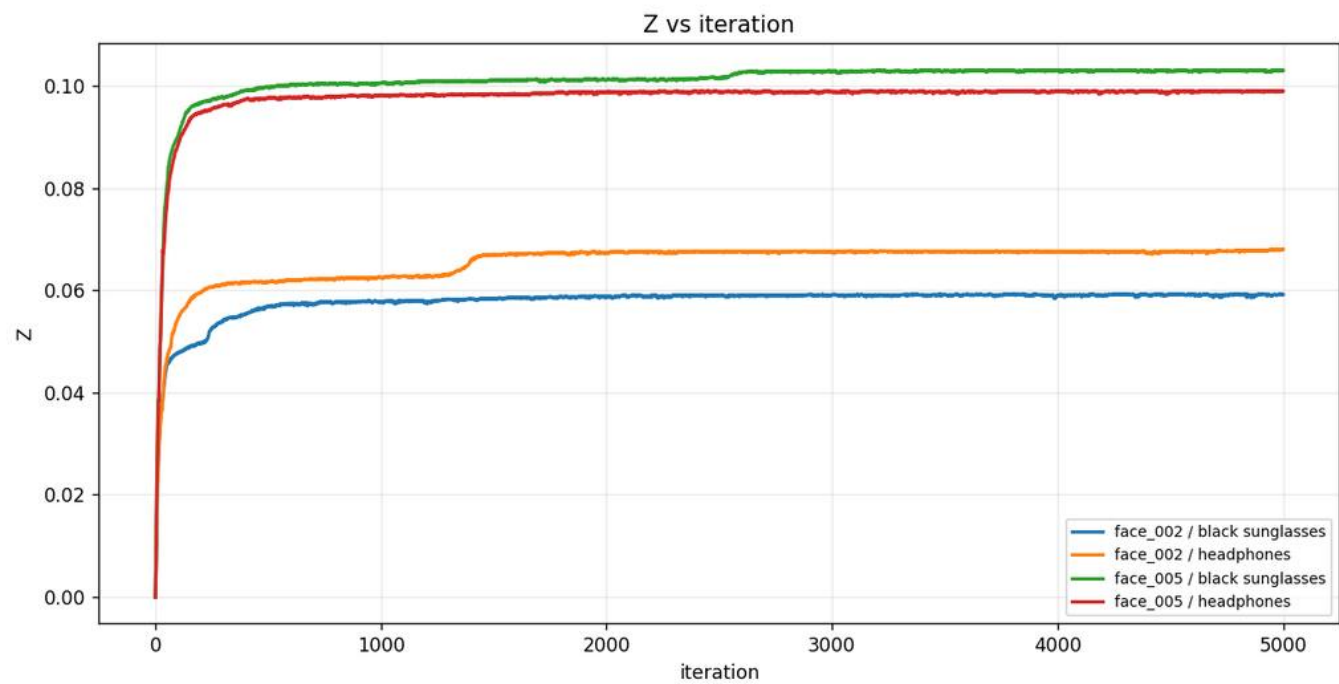


face_005 / add headphones

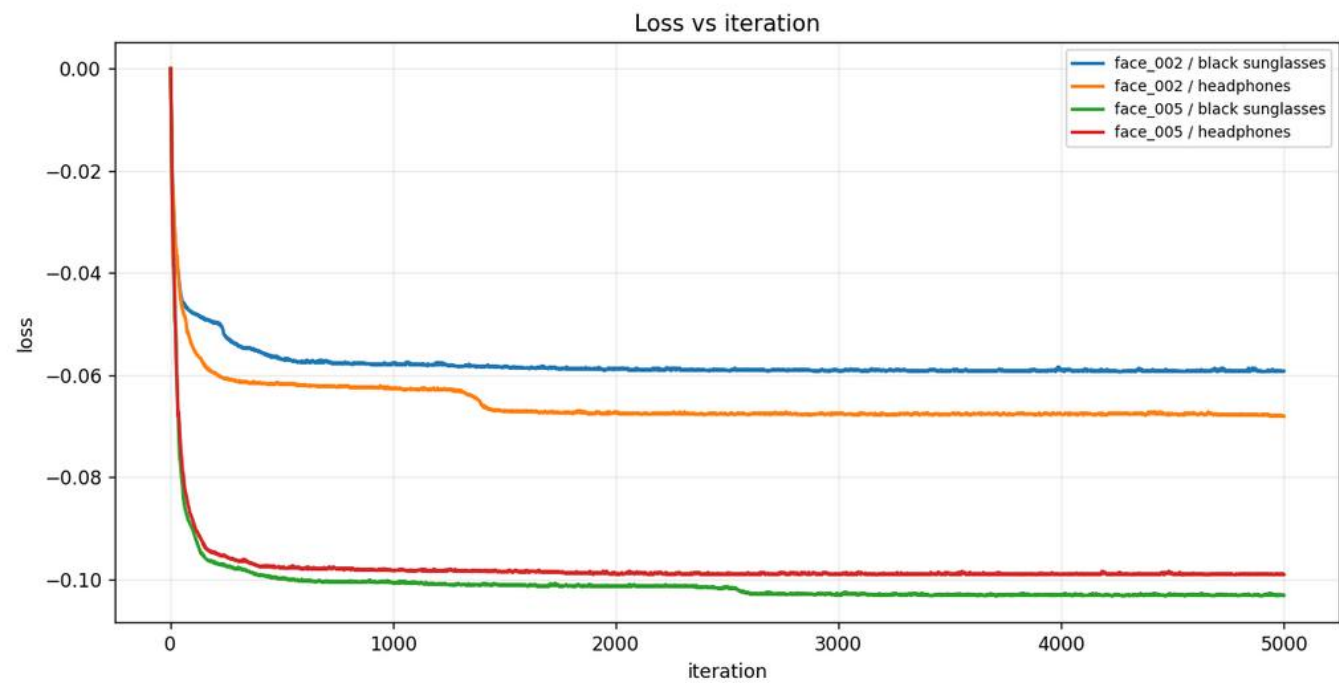


Graphs

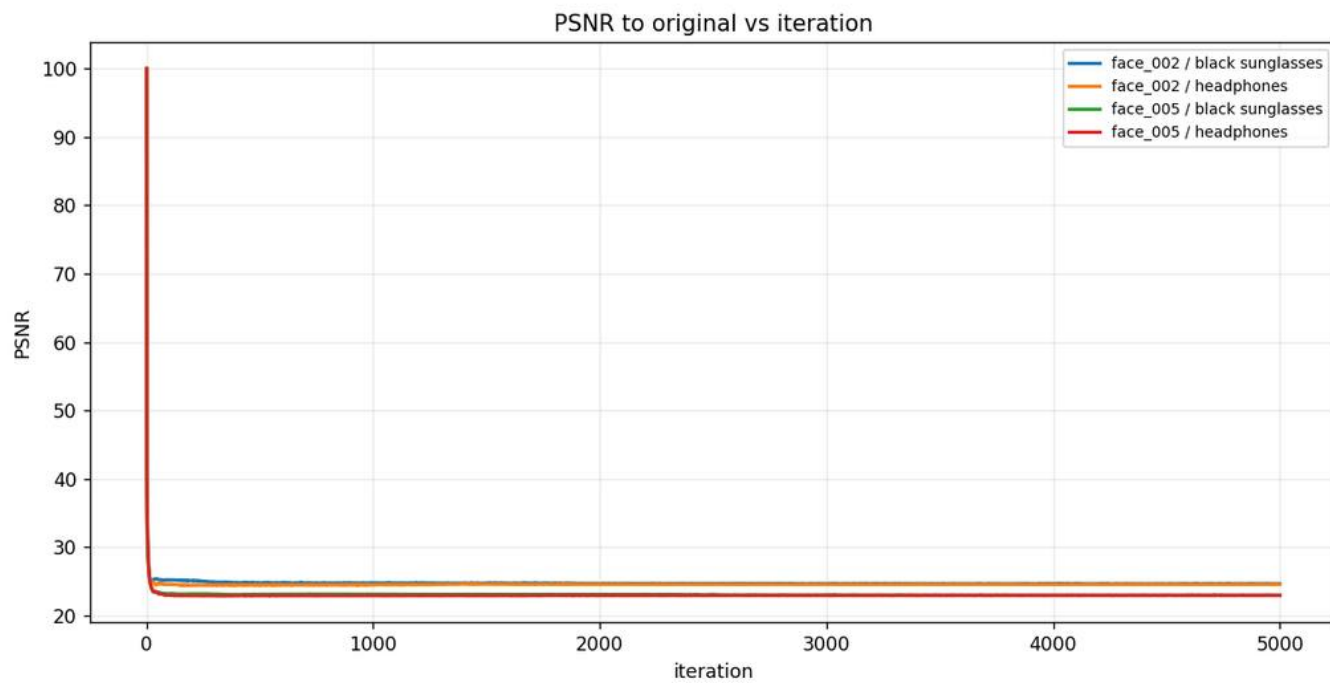
Z vs iteration



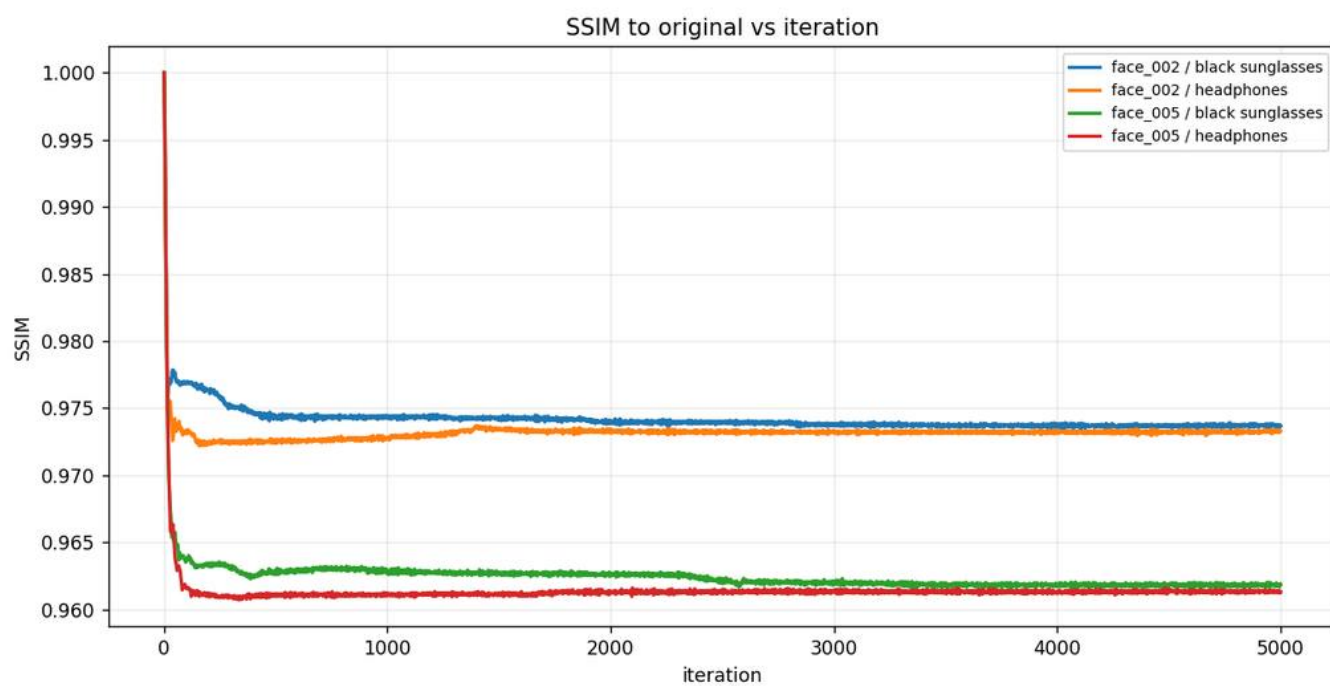
Loss vs iteration



PSNR to original vs iteration

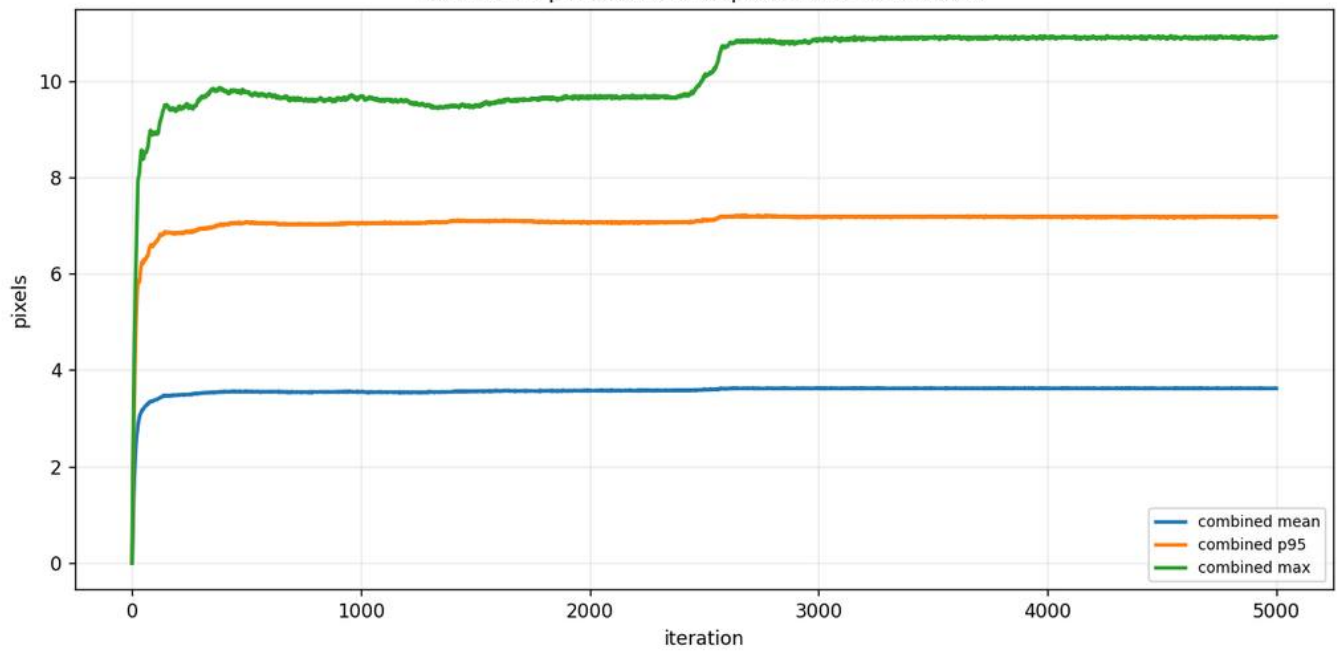


SSIM to original vs iteration



Geometric perturbation displacement vs iteration

Geometric perturbation displacement vs iteration



Geometry component max displacement vs iteration

Geometry component max displacement vs iteration

